

Algorithm Optimization

Lecture 1: Introduction to LP

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Contents

1. Introduction to linear program
2. Some applications
3. Duality of LP
4. Sensitivity and parametric analysis
5. Introduction to ILP

Linear programming

- An important optimization problem.
- Maximize (or minimize) a linear function subject to linear constraints.
- Applied in everywhere:
 - Economic (planning, scheduling,...)
 - Finance (management)
 - Control
 - Statistics
 - Transportation
 - ...

Diet problem

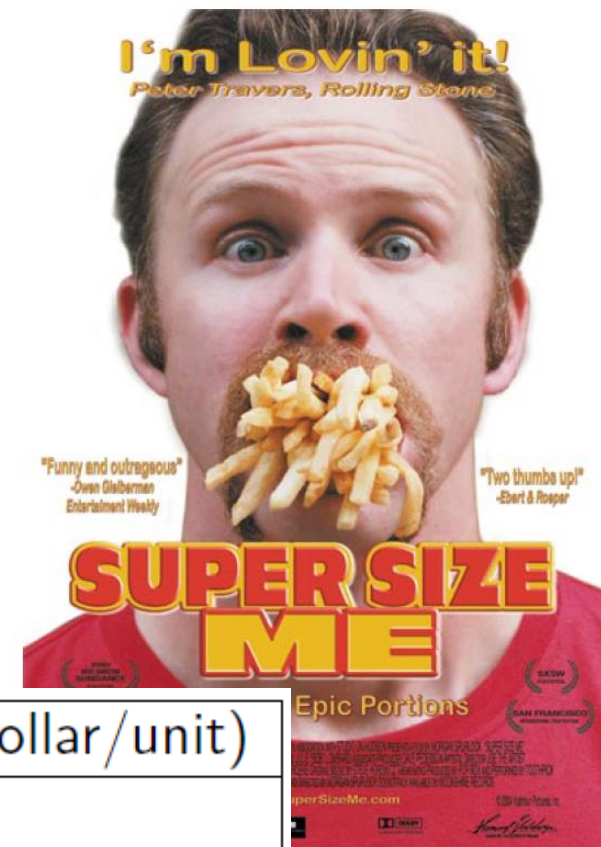
To live a healthy life, I need:

- ≥ 200 grams of carbohydrates.
- ≥ 20 grams of proteins.
- Other nutrients (ignored for now).

Food sources at grocery store:

Food	Carb/Unit	Prot/Unit	Cost (dollar/unit)
Carrots (c)	50	3	5
Rice (r)	40	1	2
Potatoes (p)	40	2	1
Quinoa (q)	40	5	10

Question: How can I satisfy my nutritional needs while paying the least amount of money possible?



Diet Problem as a Linear Program

Decision variables:

x_c	Quantity of carrots.
x_r	Quantity of rice.
x_p	Quantity of potatoes.
x_q	Quantity of quinoa.

Objective function: money paid

$$\underbrace{5}_{\$/\text{unit Carrot}} * x_c + \underbrace{2}_{\$/\text{unit Rice}} * x_r + 1 * x_p + 10 * x_q$$

Diet Problem as a Linear Program

Constraints:

1. ≥ 200 grams of carbohydrates:

$$\underbrace{50}_{\text{Carb/unit Carrot}} * x_c + 40 * x_r + 40 * x_p + 40 * x_q \geq 200 .$$

2. ≥ 20 grams of proteins:

$$3x_c + x_r + 2x_p + 5x_q \geq 20 .$$

3. Non-negative variables:

$$\begin{array}{ll} x_c \geq 0 & x_p \geq 0 \\ x_r \geq 0 & x_q \geq 0 \end{array}$$

Diet problem - cvxpy

```
import cvxpy as cp
import numpy as np
```

Optimal solution:

x =

```
c = np.array([5, 2, 1, 10])
A = np.array([[50, 40, 40, 40], [3, 1, 2, 5]])
b = np.array([200, 20])

x = cp.Variable(4)
constraint = [A@x >= b, x >= 0]
prob = cp.Problem(cp.Minimize(c.T@x), constraint)
prob.solve()

# Print result.
print("\nThe optimal value is", prob.value)
print("A solution x is")
print(x.value)
```

0
0
10.0000
0

<https://www.cvxpy.org/>

Diet problem

To live a healthy life, I need:

- ≥ 200 grams of carbohydrates.
- ≥ 20 grams of proteins.
- ≥ 10 grams of vitamins.

Food sources at grocery store:

Food	Carb/Unit	Prot/Unit	Vit./Unit	Cost (\$/unit)
Carrots (c)	20	3	10	5
Rice (r)	40	1	4	2
Potatoes (p)	40	2	1	1
Quinoa (q)	20	5	6	10

Diet problem

- Solving using cvx.

Exercise

- Solve problems 1.1 in the Vanderbei's book.
- Using cvx to solve for the solution.

A steel company must decide how to allocate next week's time on a rolling mill, which is a machine that takes unfinished slabs of steel as input and can produce either of two semi-finished products: bands and coils. The mill's two products come off the rolling line at different rates:

Bands 200 tons/hr

Coils 140 tons/hr .

They also produce different profits:

Bands \$ 25/ton

Coils \$ 30/ton .

Based on currently booked orders, the following upper bounds are placed on the amount of each product to produce:

Bands 6000 tons

Coils 4000 tons .

Given that there are 40 hours of production time available this week, the problem is to decide how many tons of bands and how many tons of coils should be produced to yield the greatest profit. Formulate this problem as a linear programming problem. Can you solve this problem by inspection?

Exercise 1.16 A manager of an oil refinery has 8 million barrels of crude oil A and 5 million barrels of crude oil B allocated for production during the coming month. These resources can be used to make either gasoline, which sells for \$38 per barrel, or home heating oil, which sells for \$33 per barrel. There are three production processes with the following characteristics:

	Process 1	Process 2	Process 3
Input crude A	3	1	5
Input crude B	5	1	3
Output gasoline	4	1	3
Output heating oil	3	1	4
Cost	\$51	\$11	\$40

All quantities are in barrels. For example, with the first process, 3 barrels of crude A and 5 barrels of crude B are used to produce 4 barrels of gasoline and 3 barrels of heating oil. The costs in this table refer to variable and allocated overhead costs, and there are no separate cost items for the cost of the crudes.

Formulate a linear programming problem that would help the manager maximize net revenue over the next month.

Outcomes for Linear Program

Three possible outcomes:

- **Infeasible:** The constraints have no solution in the first place.
- **Unbounded:** It is possible to get large values of the optimum.
- **Feasible & Bounded:** There is an optimal solution.

Examples of infeasible LP

Example 1:

$$\begin{array}{llll} \text{max.} & : & -x + 2y & \\ \text{s.t.} & & x & \leq 3 \\ & & 4y + x & \leq 7 \\ & & x & \geq 4 \\ & & 4y + x & \geq -5 \end{array}$$

Example 2:

$$\begin{array}{llll} \text{max.} & : & -x + 2y & \\ \text{s.t.} & & x + y & \geq 10 \\ & & x & \leq 8 \\ & & y & \leq 1 \end{array}$$

Show the feasible
sets using graph

Example of unbound LP

$$\begin{array}{lll} \text{max.} & : & -x + 2y \\ \text{s.t.} & & \\ & x & \leq 3 \\ & y & \geq 7 \\ & -x + 2y & \geq -5 \end{array}$$

Show the feasible set using graph.

In case of minimization, the above problem has an optimal solution.

Example

Q. Is this LP unbounded ? (use graph)

$$\begin{array}{ll}\text{maximize} & x_1 - x_2 \\ \text{subject to} & -2x_1 + x_2 \leq -1 \\ & -x_1 - 2x_2 \leq -2 \\ & x_1, x_2 \geq 0\end{array}$$

Solving Linear Programs

- Three categories of methods:
 - **Simplex** (George Dantzig, 1940s): Hill climbing search on vertices.
 - Complexity: Exponential in # Vars + # Constraints.
 - Polynomial time on almost all instances.
 - **Ellipsoid Method** (Leonid Kachiyan, 1970s): Approximate feasible region through a sequence of ellipsoids of decreasing volume.
 - Complexity: Polynomial time (not a practical method).

Solving Linear Programs

- Interior Point Methods (Narendra Karmarkar, 1982): Practical polynomial time algorithm.
- Other approaches for special cases:
 - Two variables per constraint case.
 - Transshipment problem, problems over graphs: Network Simplex Method.
 - ...

Standard LP problem

$$\begin{array}{ll}\text{maximize} & c_1x_1 + c_2x_2 + \cdots + c_nx_n \\ \text{subject to} & a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1 \\ & a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2 \\ & \vdots \\ & a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m \\ & x_1, x_2, \dots, x_n \geq 0.\end{array}$$

Other standard form of LP problem

- The other books can derive the theory using other standard forms. E.g., Bertsimas' book uses the following form

$$\begin{array}{ll}\text{minimize} & \mathbf{c}'\mathbf{x} \\ \text{subject to} & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0},\end{array}$$

Transform the LP problem

- Free variables:
 - replace x_i by $x_i^+ - x_i^-$, where $x_i^+ \geq 0$ and $x_i^- \geq 0$
- Slack variables s_i :
 - $a_i^T x \leq b_i$ equivalent to $a_i^T x + s_i = b_i$, where $s_i \geq 0$
 - The surplus variable s_i also is used to transform $a_i^T x \geq b_i$ to $a_i^T x - s_i = b_i$

Example

- Given problem:

$$\begin{array}{ll}\text{minimize} & 2x_1 + 4x_2 \\ \text{subject to} & x_1 + x_2 \geq 3 \\ & 3x_1 + 2x_2 = 14 \\ & x_1 \geq 0,\end{array}$$

- The equivalent problem is:

$$\begin{array}{ll}\text{minimize} & 2x_1 + 4x_2^+ - 4x_2^- \\ \text{subject to} & x_1 + x_2^+ - x_2^- - x_3 = 3 \\ & 3x_1 + 2x_2^+ - 2x_2^- = 14 \\ & x_1, x_2^+, x_2^-, x_3 \geq 0.\end{array}$$

Transform the LP problem – piecewise linear

- Piecewise linear function:

$$\max_{i=1,\dots,m} (c_i^T x + d_i)$$

- The problem

$$\begin{aligned} &\text{minimize} && \max_{i=1,\dots,m} (\mathbf{c}'_i \mathbf{x} + d_i) \\ &\text{subject to} && \mathbf{Ax} \geq \mathbf{b}. \end{aligned}$$

equivalents to the following LP prob.:

$$\begin{aligned} &\text{minimize} && z \\ &\text{subject to} && z \geq \mathbf{c}'_i \mathbf{x} + d_i, && i = 1, \dots, m, \\ &&& \mathbf{Ax} \geq \mathbf{b}, \end{aligned}$$

Transform the LP problem - absolute values

- The problem with $c_i \geq 0$

$$\begin{array}{ll}\text{minimize} & \sum_{i=1}^n c_i |x_i| \\ \text{subject to} & \mathbf{Ax} \geq \mathbf{b},\end{array}$$

equivalents to

$$\begin{array}{ll}\text{minimize} & \sum_{i=1}^n c_i z_i \\ \text{subject to} & \mathbf{Ax} \geq \mathbf{b} \\ & x_i \leq z_i, & i = 1, \dots, n, \\ & -x_i \leq z_i, & i = 1, \dots, n.\end{array}$$

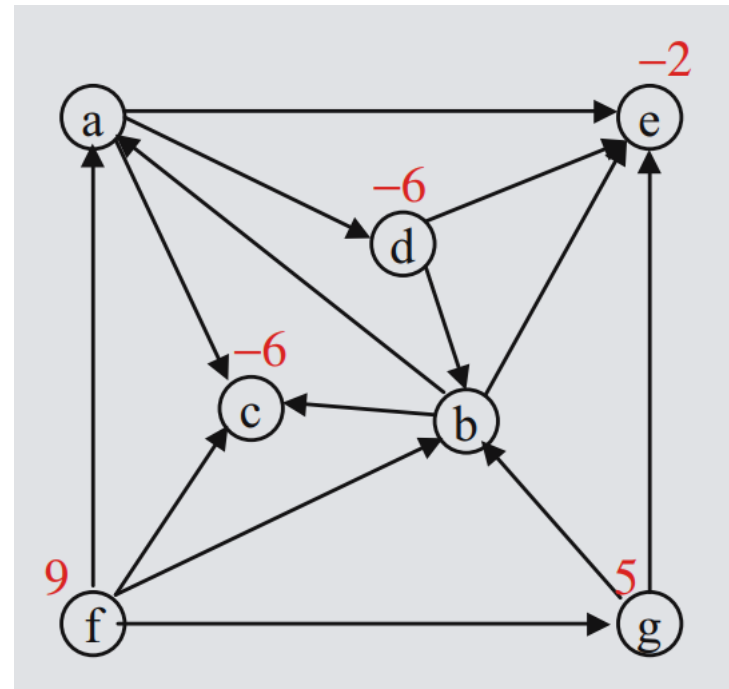
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Minimum-cost network flow problem

Set of nodes and arcs $G(N, A)$.

E.g., a network having 7 nodes (N) and 14 arcs (A). The red numbers denote the supply or demand at the node (b_i)

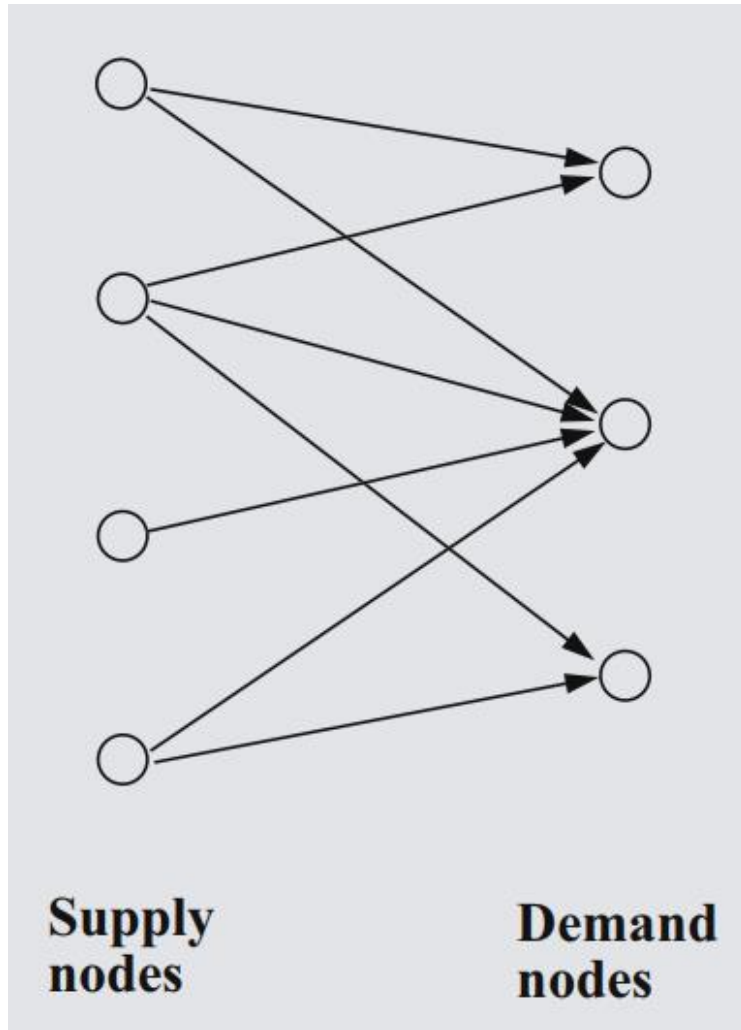


Minimum-cost network flow problem

- c_{ij} : the cost of shipping one unit from i to j directly
- x_{ij} : quantity shipped directly from i to j
- Objective:
$$\text{minimize } \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}.$$
- Flow balance constraints:

$$\sum_{\substack{i: \\ (i,k) \in \mathcal{A}}} x_{ik} - \sum_{\substack{j: \\ (k,j) \in \mathcal{A}}} x_{kj} = -b_k, \quad k \in \mathcal{N}.$$

Transportation problem



- S : set of supply nodes
- D : set of demand nodes
- $r_i, i \in S$: supplies at supply node i
- $s_i, i \in D$: demands at demand node i .
- Assumption: total of supplies = total of demands

Bipartite graph

Transportation problem

$$\begin{array}{ll}\text{minimize} & \sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{D}} c_{ij} x_{ij} \\ \text{subject to} & \sum_{j \in \mathcal{D}} x_{ij} = r_i \quad i \in \mathcal{S} \\ & \sum_{i \in \mathcal{S}} x_{ij} = s_j \quad j \in \mathcal{D} \\ & x_{ij} \geq 0 \quad i \in \mathcal{S}, j \in \mathcal{D}.\end{array}$$

Assignment Problem

- Set S of m people, a set D of m tasks
- Cost c_{ij} associated with assigning person i to task j
- Let's
$$x_{ij} = \begin{cases} 1 & \text{if person } i \text{ is assigned task } j, \\ 0 & \text{otherwise,} \end{cases}$$
- Each person is assigned exactly one task
- Every task gets covered by someone

Assignment Problem

$$\text{minimize } \sum_{i \in \mathcal{S}} \sum_{j \in \mathcal{D}} c_{ij} x_{ij}.$$

$$\text{s.t. } \sum_{j \in \mathcal{D}} x_{ij} = 1, \quad \text{for all } i \in \mathcal{S}.$$

$$\sum_{i \in \mathcal{S}} x_{ij} = 1, \quad \text{for all } j \in \mathcal{D}.$$

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Linear program

$$\begin{array}{ll} \text{maximize} & c_1x_1 + \cdots + c_nx_n \\ \text{s.t.} & a_{j1}x_1 + \cdots + a_{jn}x_n \leq b_j \quad j \in \{1, 2, \dots, m\} \\ & x_1, \dots, x_n \geq 0 \end{array}$$

Example

$$\begin{array}{llllll} \text{maximize} & x_1 & + & 2x_2 & & \\ \text{s.t.} & x_1 & & & \leq & 3 \\ & & & x_2 & \leq & 3 \\ & -x_1 & + & x_2 & \leq & 1 \\ & x_1 & + & x_2 & \leq & 5 \\ & & & x_1, x_2 & \geq & 0 \end{array}$$

Example

Q: Finding the upper bound on the solution?

$$\begin{array}{llllll} \text{maximize} & x_1 & + & 2x_2 & & \\ \text{s.t.} & x_1 & & & \leq & 3 \\ & & & x_2 & \leq & 3 \\ & -x_1 & + & x_2 & \leq & 1 \\ & x_1 & + & x_2 & \leq & 5 \\ & & & x_1, x_2 & \geq & 0 \end{array}$$

Upper Bound

$$\begin{array}{llllll} \text{maximize} & x_1 & + & 2x_2 & & \leftarrow z \\ \text{s.t.} & x_1 & & & \leq & 3 \leftarrow \mathbf{C1} \\ & & & x_2 & \leq & 3 \leftarrow \mathbf{C2} \\ & -x_1 & + & x_2 & \leq & 1 \leftarrow \mathbf{C3} \\ & x_1 & + & x_2 & \leq & 5 \leftarrow \mathbf{C4} \\ & & & x_1, x_2 & \geq & 0 \end{array}$$

Using the inequality above what can you say about the upper bound of objective?

$$C1 + 2 \cdot C2 \Rightarrow z \leq 9.$$

$$2 \cdot C4 \Rightarrow z \leq 10.$$

Question #1: $-1 \cdot C3 + 3 \cdot C2 \Rightarrow z \leq ?$

Question #2: $2 \cdot C3 + C1 \Rightarrow z \leq ?$

Upper Bound

$$\begin{array}{llllllll}
 \text{maximize} & c_1x_1 & + & \cdots & + & c_nx_n & & \mathbf{z} \\
 \text{s.t.} & a_{11}x_1 & + & \cdots & + & a_{1n}x_n & \leq & b_1 \leftarrow \mathbf{C1} \\
 & a_{21}x_1 & + & \cdots & + & a_{2n}x_n & \leq & b_2 \leftarrow \mathbf{C2} \\
 & & & \ddots & & & \vdots & \\
 & a_{m1}x_1 & & \cdots & + & a_{mn}x_n & \leq & b_m \leftarrow \mathbf{Cm} \\
 & & & & & x_1, \dots, x_n & \geq & 0
 \end{array}$$

Consider $y_1C_1 + y_2C_2 + \cdots + y_mC_m$.

Q: What are the conditions on $y_1, \dots, y_m$ so that this combination upper bounds z ?

Upper Bound

Conditions:

$$\begin{array}{rcl} a_{11}y_1 + a_{21}y_2 + \cdots + a_{m1}y_m & \geq & c_1 \\ a_{12}y_1 + a_{22}y_2 + \cdots + a_{m2}y_m & \geq & c_2 \\ & \vdots & \\ a_{1n}y_1 + a_{2n}y_2 + \cdots + a_{mn}y_m & \geq & c_n \\ y_1, y_2, \dots, y_m & \geq & 0 \end{array}$$

Upper Bound: $b_1y_1 + \cdots + b_my_m$

Q: What values of $y_1, \dots, y_m$ can yield best upper bound of the solution?

Best upper bound on LP Solution

$$\begin{array}{ll} \text{minimize} & b_1y_1 + b_2y_2 + \cdots + b_my_m \\ \text{s.t.} & a_{11}y_1 + a_{21}y_2 + \cdots + a_{m1}y_m \geq c_1 \\ & a_{12}y_1 + a_{22}y_2 + \cdots + a_{m2}y_m \geq c_2 \\ & \vdots \\ & a_{1n}y_1 + a_{2n}y_2 + \cdots + a_{mn}y_m \geq c_n \\ & y_1, y_2, \dots, y_m \geq 0 \end{array}$$

This is called the dual problem.

Example 1: Primal with inequality constraint

Primal Problem: decision variables are $x_1, \dots, x_n$.

$$\begin{array}{ll} \text{maximize} & c_1 x_1 + \dots + c_n x_n \\ \text{s.t.} & a_{j1} x_1 + \dots + a_{jn} x_n \leq b_j \quad j \in \{1, 2, \dots, m\} \\ & x_1, \dots, x_n \geq 0 \end{array}$$

Dual Problem: decision variables are $y_1, \dots, y_m$.

$$\begin{array}{ll} \text{minimize} & b_1 y_1 + \dots + b_m y_m \\ \text{s.t.} & a_{1j} y_1 + \dots + a_{mj} y_m \geq c_j \quad j \in \{1, 2, \dots, n\} \\ & y_1, \dots, y_m \geq 0 \end{array}$$

Forming dual problem using Lagrange method

- Lagrange function (the upper bound):

$$\begin{aligned} g(x, y, z) &= c^T x - y^T (Ax - b) + z^T x \\ &= (c^T - y^T A + z^T)x + y^T b \end{aligned}$$

z and y are Lagrange multipliers

- For $g(x, y, z)$ to be an upper bound of the LP solution, we need $z, y \geq 0$. Hence, $y^T (Ax - b) \leq 0$ and $z^T x \geq 0$.
- $\max_x g(x, y, z) = y^T b$ if $c^T - y^T A + z^T = 0$;
otherwise, $\max_x g(x, y, z) = \infty$

Dual problem

- Dual problem is to find the best upper bound:

$$\begin{aligned} \min & b^T y \\ \text{s.t.} & c^T - y^T A + z^T = 0 \\ & y, z \geq 0 \end{aligned}$$

- Equivalently,

$$\begin{aligned} \min & b^T y \\ \text{st.} & c \leq A^T y \\ & y \geq 0 \end{aligned}$$

Example 2: Primal with equality constraint

$$\begin{array}{ll}\text{maximize} & \sum_{j=1}^n c_j x_j \\ \text{subject to} & \sum_{j=1}^n a_{ij} x_j = b_i \quad i = 1, 2, \dots, m \\ & x_j \geq 0 \quad j = 1, 2, \dots, n.\end{array}$$

$$\begin{array}{ll}\text{minimize} & \sum_{i=1}^m b_i y_i \\ \text{subject to} & \sum_{i=1}^m y_i a_{ij} \geq c_j \quad j = 1, 2, \dots, n.\end{array}$$

Forming dual problem using Lagrange method

- Lagrange function:

$$\begin{aligned} g(x, y, z) &= c^T x - y^T (Ax - b) + z^T x \\ &= (c^T - y^T A + z^T)x + y^T b \end{aligned}$$

- For $g(x, y, z)$ to be an upper bound of the LP solution, we need $z \geq 0$ (Here, we don't need constraint $y \geq 0$) $\Rightarrow y^T (Ax - b) = 0$ and $z^T x \geq 0$.
- $\max_x g(x, y, z) = y^T b$ if $c^T - y^T A + z^T = 0$;
otherwise, $\max_x g(x, y, z) = \infty$

Dual problem

- Dual problem is to find the best upper bound:

$$\begin{aligned} \min \quad & b^T y \\ \text{s.t.} \quad & c^T - y^T A + z^T = 0 \\ & z \geq 0 \end{aligned}$$

- Equivalently,

$$\begin{aligned} \min \quad & b^T y \\ \text{st.} \quad & A^T y \geq c \end{aligned}$$

Rules for forming the dual of maximization LP problem

Primal	Dual
Equality constraint	Free variable
Inequality constraint	Nonnegative variable
Free variable	Equality constraint
Nonnegative variable	Inequality constraint

Equality constraint $\Leftrightarrow$ free variable

Inequality constraint $\Leftrightarrow$ non-negative variable

Quiz

Find the dual problem of the following LP:

$$\begin{aligned} &\text{maximize } 2x_1 + 8x_2 - x_3 - 2x_4 \\ &\text{subject to } 2x_1 + 3x_2 + 6x_4 \leq 6 \\ &\quad \quad \quad -2x_1 + 4x_2 + 3x_3 \leq 1.5 \\ &\quad \quad \quad 3x_1 + 2x_2 - 2x_3 - x_4 \leq 4 \\ &\quad \quad \quad x_1, x_2, x_3, x_4 \geq 0. \end{aligned}$$

Find the dual of the dual problem.

Weak duality theorem

Given an LP in standard form:

$$\begin{array}{ll} \text{maximize} & c_1x_1 + \cdots + c_nx_n \\ \text{s.t.} & a_{j1}x_1 + \cdots + a_{jn}x_n \leq b_j \quad j \in \{1, 2, \dots, m\} \\ & x_1, \dots, x_n \geq 0 \end{array}$$

THEOREM 5.1. *If $(x_1, x_2, \dots, x_n)$ is feasible for the primal and $(y_1, y_2, \dots, y_m)$ is feasible for the dual, then*

$$\sum_j c_j x_j \leq \sum_i b_i y_i.$$

Strong duality theorem

THEOREM 5.2. *If the primal problem has an optimal solution,*

$$x^* = (x_1^*, x_2^*, \dots, x_n^*),$$

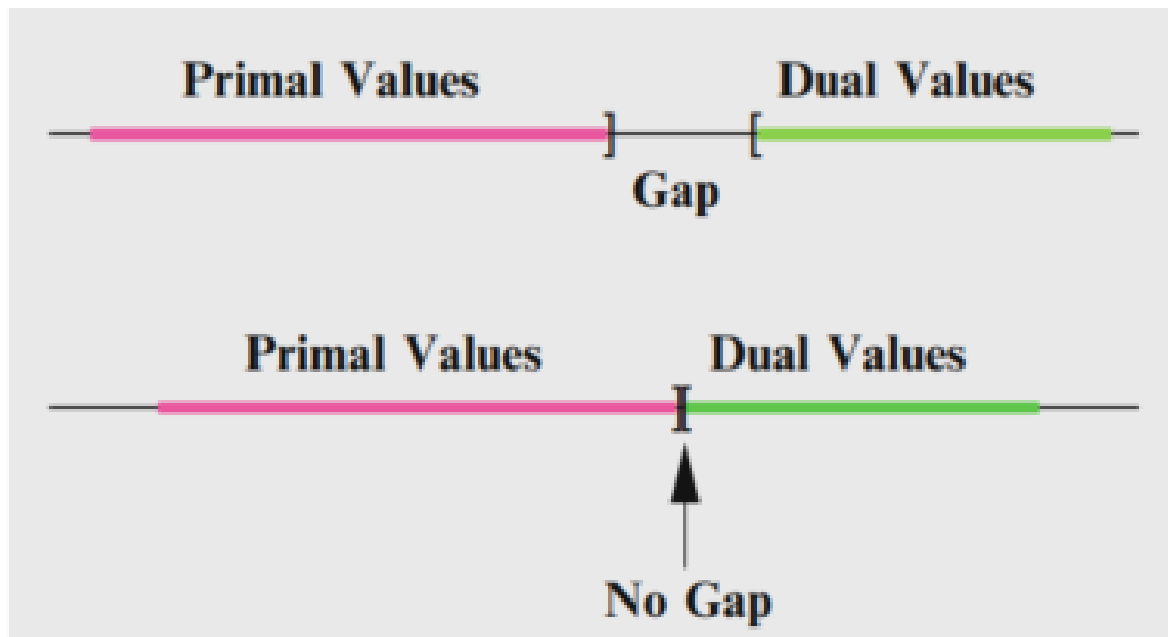
then the dual also has an optimal solution,

$$y^* = (y_1^*, y_2^*, \dots, y_m^*),$$

such that

$$(5.2) \quad \sum_j c_j x_j^* = \sum_i b_i y_i^*.$$

Duality gap



Complementary slackness

Primal Problem:

$$\begin{aligned} \max \quad & \sum_{j=1}^n c_j x_j \\ \text{s.t.} \quad & \sum_{j=1}^n a_{ij} x_j + w_i = b_i \quad i = 1, \dots, m \\ & x_j \geq 0 \quad j = 1, \dots, n \\ & w_i \geq 0 \quad i = 1, \dots, m \end{aligned}$$

Dual Problem:

$$\begin{aligned} \min \quad & \sum_{i=1}^m b_i y_i \\ \text{s.t.} \quad & \sum_{i=1}^m y_i a_{ij} - z_j = c_j \quad j = 1, \dots, n \\ & y_i \geq 0 \quad i = 1, \dots, m \\ & z_j \geq 0 \quad j = 1, \dots, n \end{aligned}$$

Theorem. *At optimality, we have*

$$\begin{aligned} x_j z_j &= 0, & \text{for } j = 1, 2, \dots, n, \\ w_i y_i &= 0, & \text{for } i = 1, 2, \dots, m. \end{aligned}$$

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Sensitivity

$$\begin{array}{llll} \text{max.} & -2x_1 - 5x_2 - 15x_3 & & \\ & -20x_1 - x_2 - x_3 & \leq & -200 \quad \leftarrow \text{CARBS}(0.07) \\ & -x_1 - 30x_2 - 40x_3 & \leq & -50 \quad \leftarrow \text{PROT}(0) \\ & -x_1 - 10x_2 - 5x_3 & \leq & -40 \quad \leftarrow \text{VIT}(0.5) \\ & x_1 & \leq & 20 \quad \leftarrow \text{LIM FX } 0 \\ & x_2 & \leq & 10 \quad \leftarrow \text{LIM FY } 0 \\ & x_3 & \leq & 5 \quad \leftarrow \text{LIM FZ } 0 \\ & x_1, x_2, x_3 & \geq & 0 \end{array}$$

Experiment: Play around with RHS of inequalities and observe solution.

Sensitivity

$$\begin{array}{llll} \text{max.} & -2x_1 - 5x_2 - 15x_3 & & \\ & -20x_1 - x_2 - x_3 & \leq & -200 \quad \leftarrow \text{CARBS}(0.07) \\ & -x_1 - 30x_2 - 40x_3 & \leq & -50 \quad \leftarrow \text{PROT}(0) \\ & -x_1 - 10x_2 - 5x_3 & \leq & -40 \quad \leftarrow \text{VIT}(0.5) \\ & x_1 & \leq & 20 \quad \leftarrow \text{LIM FX } 0 \\ & x_2 & \leq & 10 \quad \leftarrow \text{LIM FY } 0 \\ & x_3 & \leq & 5 \quad \leftarrow \text{LIM FZ } 0 \\ & x_1, x_2, x_3 & \geq & 0 \end{array}$$

Q1: If dual variable corresponding to a constraint is 0 does that mean that we can drop the constraint and still get the same answer?

A1: YES.

Q2: Can we, then, change the RHS of the inequality to arbitrary values?

A2: NO.

Sensitivity Theorem

Consider **Primal Problem**: decision variables are $x_1, \dots, x_n$.

$$\begin{array}{ll} \text{maximize} & c_1x_1 + \dots + c_nx_n \\ \text{s.t.} & a_{j1}x_1 + \dots + a_{jn}x_n \leq b_j \quad j \in \{1, 2, \dots, m\} \\ & x_1, \dots, x_n \geq 0 \end{array}$$

Optimal Value: z^* .

Primal Optimal: $x_1^*, \dots, x_n^*$. **Dual Optimal**: $y_1^*, \dots, y_m^*$.

Modified LP:

$$\begin{array}{ll} \text{maximize} & c_1x_1 + \dots + c_nx_n \\ \text{s.t.} & a_{j1}x_1 + \dots + a_{jn}x_n \leq b_j + t_j \quad j \in \{1, 2, \dots, m\} \\ & x_1, \dots, x_n \geq 0 \end{array}$$

Optimal Value: There is $\epsilon > 0$ s.t., if $|t_i| \leq \epsilon$, $z' = z^* + \sum_i t_i y_i$.

Shadow Cost: The dual decision variables are interpreted as “shadow cost”.

Problem 1

Consider the LP problem:

$$\begin{array}{ll}\text{minimize} & 13x_1 + 10x_2 + 6x_3 \\ \text{subject to} & 5x_1 + x_2 + 3x_3 = 8 \\ & 3x_1 + x_2 = 3 \\ & x_1, x_2, x_3 \geq 0,\end{array}$$

- Write the dual problem
- Write the complementary slackness

Quiz - answer

- The LP problem and its dual:

minimize	$13x_1 + 10x_2 + 6x_3$	maximize	$8p_1 + 3p_2$
subject to	$5x_1 + x_2 + 3x_3 = 8$	subject to	$5p_1 + 3p_2 \leq 13$
	$3x_1 + x_2 = 3$		$p_1 + p_2 \leq 10$
	$x_1, x_2, x_3 \geq 0,$		$3p_1 \leq 6.$

Exercises

Problem 2 (Problem 4.8-d, Boyd et.al.) Minimizing a linear function.

$$\begin{aligned} & \text{minimize } c^T x \\ & \text{subject to } 1^T x = 1, x \geq 0 \end{aligned}$$

Problem 3 (Problem 4.8-f, Boyd et.al.) Minimizing a linear function over a unit box with a weighted budget constraint. (fractional Knapsack problem)

$$\begin{aligned} & \text{minimize } c^T x \\ & \text{subject to } d^T x = \alpha, 0 \leq x \leq 1 \\ & \text{with } d > 0, \text{ and } 0 \leq \alpha \leq 1^T d. \end{aligned}$$

Problem 3 (Problems 5.5-a, page 80; Vanderbei)

Consider the following linear programming problem:

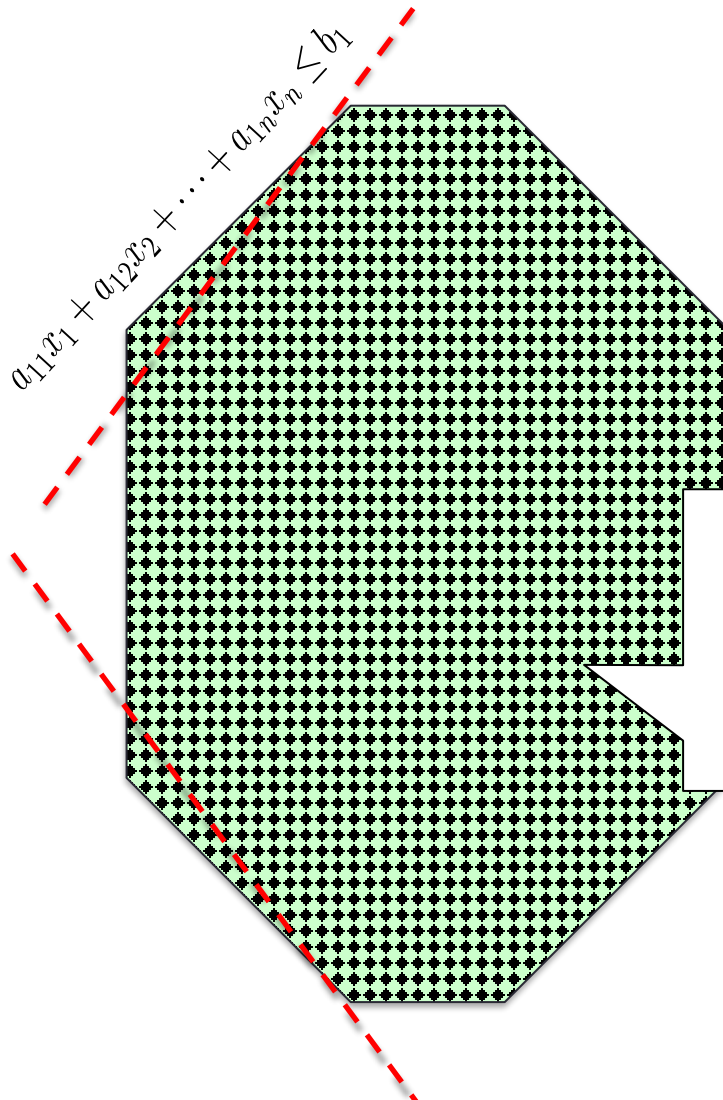
$$\begin{array}{ll}\text{maximize} & 2x_1 + 8x_2 - x_3 - 2x_4 \\ \text{subject to} & 2x_1 + 3x_2 \quad \quad + 6x_4 \leq 6 \\ & -2x_1 + 4x_2 + 3x_3 \quad \leq 1.5 \\ & 3x_1 + 2x_2 - 2x_3 - 4x_4 \leq 4 \\ & x_1, x_2, x_3, x_4 \geq 0.\end{array}$$

- (a) Write down the dual problem.
- (b) Write the complementary slackness property showing the relations between primal and dual solutions?

Contents

1. Introduction to linear program
2. Some applications
3. Duality of LP
4. Sensitivity and parametric analysis
- 5. Introduction to ILP**

Integer Linear Programming (ILP)



$$\begin{array}{llllll} \max & c_1x_1 & +c_2x_2 & +\cdots+ & c_nx_n & \\ \text{s.t.} & a_{11}x_1 & +a_{12}x_2 & +\cdots+ & a_{1n}x_n & \leq b_1 \\ & & & & \vdots & \\ & a_{m1}x_1 & +a_{m2}x_2 & +\cdots+ & a_{mn}x_n & \leq b_m \\ & \color{red}{x_1, \dots, x_n \in \mathbb{Z}} & & & & \end{array}$$

Feasible Region: Z-Polyhedron
(n dimensional)

Integer Linear Program

- Feasibility of ILP:
 - *Integer feasible solution.*

$$\begin{array}{llllll} \max & c_1x_1 & +c_2x_2 & +\cdots+ & c_nx_n & \\ \text{s.t.} & a_{11}x_1 & +a_{12}x_2 & +\cdots+ & a_{1n}x_n & \leq b_1 \\ & & & & \ddots & \vdots \\ & a_{m1}x_1 & +a_{m2}x_2 & +\cdots+ & a_{mn}x_n & \leq b_m \end{array}$$

$x_1, \dots, x_n \in \mathbb{Z}$

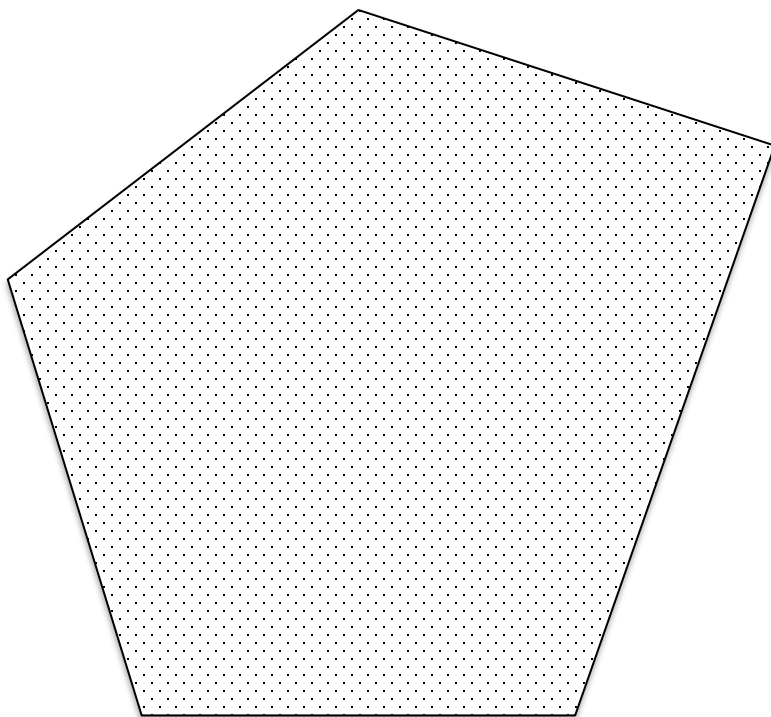
- Unbounded ILP:
 - **Integer feasible solutions** can achieve arbitrarily large values for the objective.

Linear Programming Relaxation

$$\begin{array}{llllll} \max & c_1x_1 & +c_2x_2 & +\cdots+ & c_nx_n & \\ \text{s.t.} & a_{11}x_1 & +a_{12}x_2 & +\cdots+ & a_{1n}x_n & \leq b_1 \\ & & & & & \vdots \\ & a_{m1}x_1 & +a_{m2}x_2 & +\cdots+ & a_{mn}x_n & \leq b_m \\ & x_1, \dots, x_n & \in & \mathbb{Z} & & \end{array}$$

Q: What happens to the answer if we take away the integrality constraints?

Feasible Regions

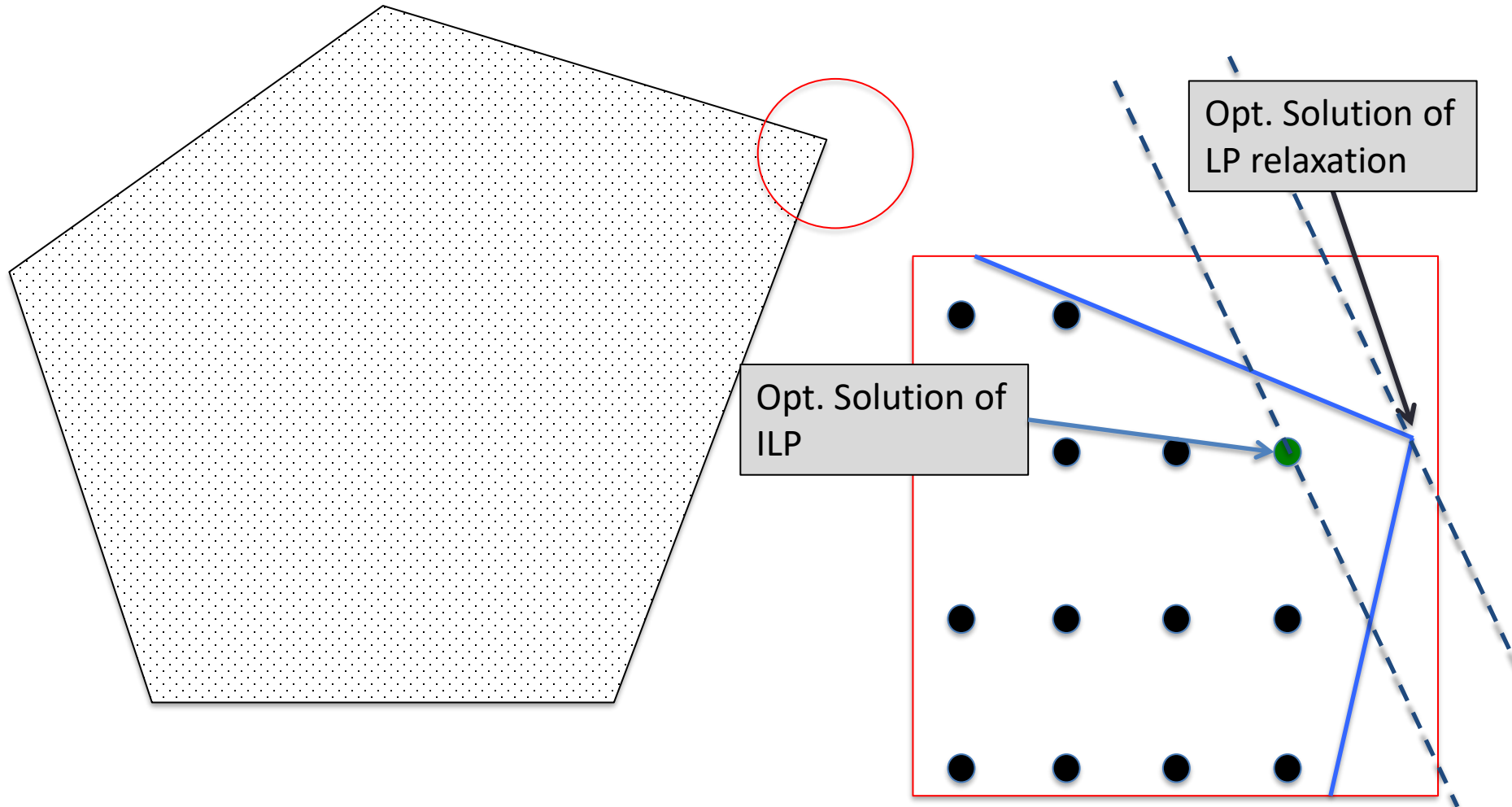


$$\begin{array}{llllll} \max & c_1x_1 & +c_2x_2 & +\cdots+ & c_nx_n & \\ \text{s.t.} & a_{11}x_1 & +a_{12}x_2 & +\cdots+ & a_{1n}x_n & \leq b_1 \\ & & & & \ddots & \vdots \\ & a_{m1}x_1 & +a_{m2}x_2 & +\cdots+ & a_{mn}x_n & \leq b_m \end{array}$$

$$x_1, \dots, x_n \in \mathbb{Z}$$

ILP feasible region is a
subset of LP feasible region

Case-1: Both LP and ILP are feasible.

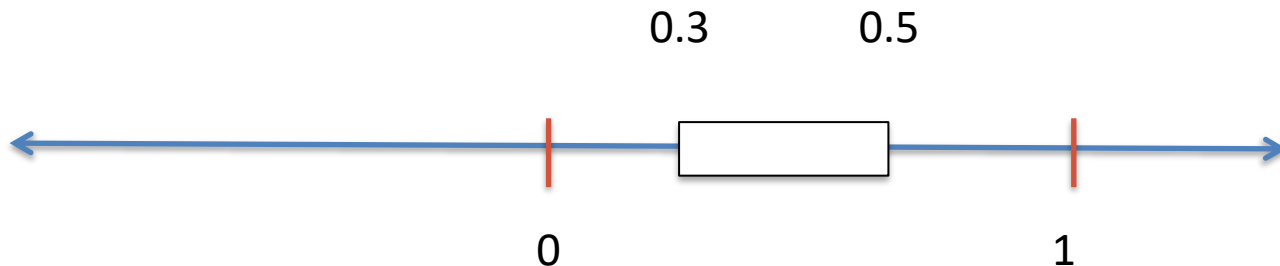


Case-2: LP relaxation is feasible, ILP is infeasible.

$$\begin{array}{ll}\max & x \\ \text{s.t.} & \\ & 3 \leq 10x \leq 5\end{array}$$

ILP is infeasible.

LP relaxation has optimal solution: 0.5



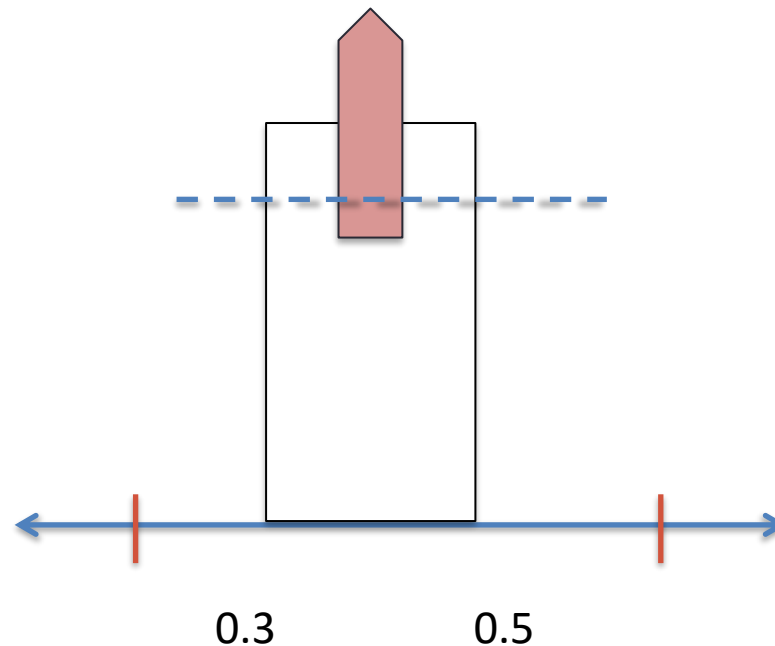
Case 3: ILP is infeasible, LP is unbounded.

Example:

$$\begin{array}{ll}\max & y \\ 3 \leq & 10x \leq 5 \\ 0 \leq & y\end{array}$$

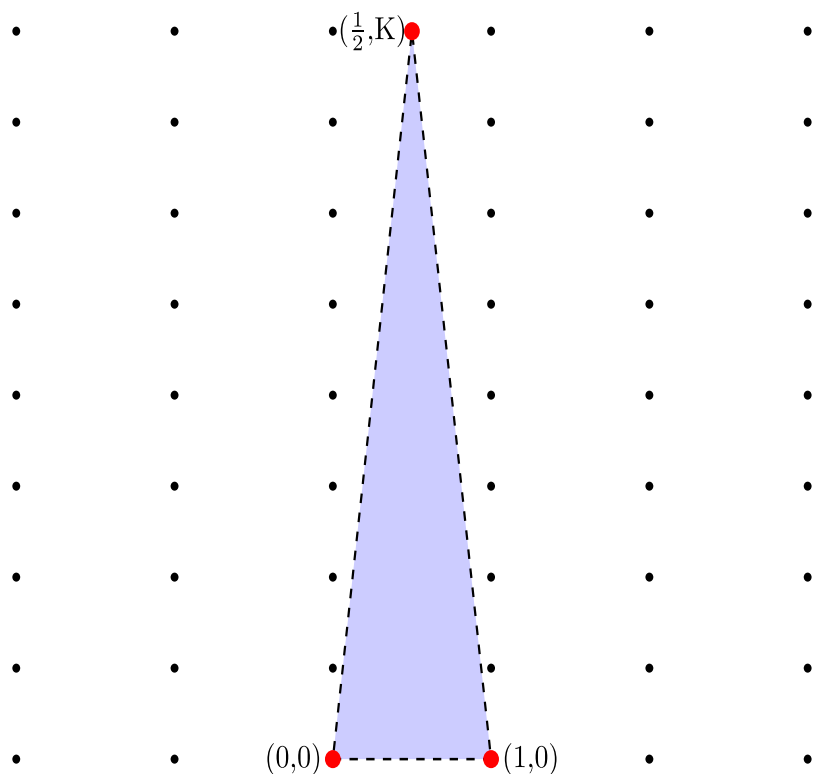
ILP is infeasible.

LP relaxation is unbounded



Claim

LP relaxation's answer can be arbitrarily larger than the ILP's answer.



max

x_2

s.t

$x_2 \geq 0$

$2Kx_1 - x_2 \geq 0$

$-2Kx_1 - x_2 \geq -2K$

$x_1, x_2 \in \mathbb{Z}$

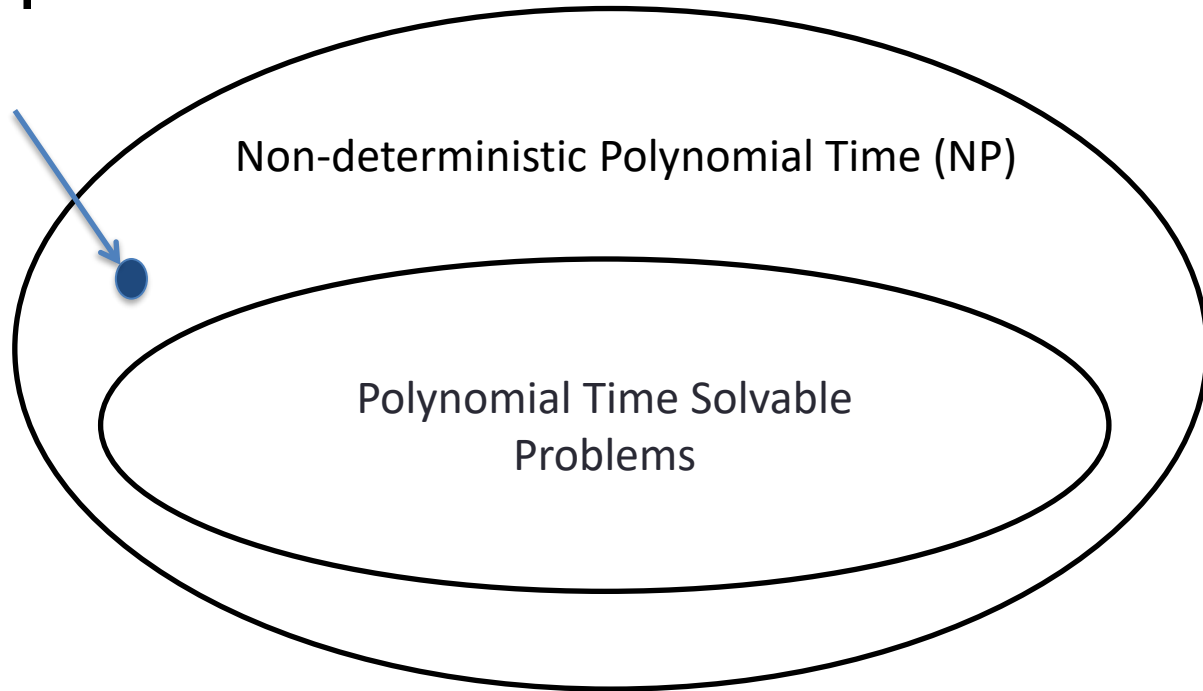
Integer Linear Programming

- Complexity of Integer Linear Programs
 - NP-Completeness
 - Some special cases of ILPs.

Complexity of Integer Linear Programs

Integer Linear Programming problems are NP-complete

Integer Linear Programming



Implications of P vs. NP question

- $P=NP$
 - Considered an unlikely possibility by experts.
 - In this case, we will be able to solve ILPs in polynomial time.
- $P \neq NP$
 - In this case, we can show a non-polynomial lower bound on the complexity of solving ILPs.

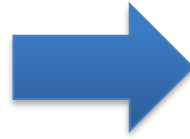
3-SAT Problem

x_1, x_2, x_3, x_4 Boolean Variables

$(x_1 \text{ OR } x_2 \text{ OR } \neg x_3)$

$(\neg x_2 \text{ OR } \neg x_4 \text{ OR } x_1)$

$(x_1 \text{ OR } x_2 \text{ OR } \neg x_3)$



Find values for Boolean variables such that

All the Clauses are True.

Reducing 3-SAT to ILP

$x_1, \dots, x_n$ are Boolean variables.

$$C_1 : (\ell_{1,1} \text{ OR } \ell_{1,2} \text{ OR } \ell_{1,3})$$

$$\vdots$$
$$\ddots$$

m Clauses.

$$C_m : (\ell_{m,1} \text{ OR } \ell_{m,2} \text{ OR } \ell_{m,3})$$

$\ell_{i,j}$ stands for a variable x_k or its negation $\neg x_k$

ILP reduction.

$$x_j \rightarrow y_j \in \{0, 1\}$$

False = 0
True = 1

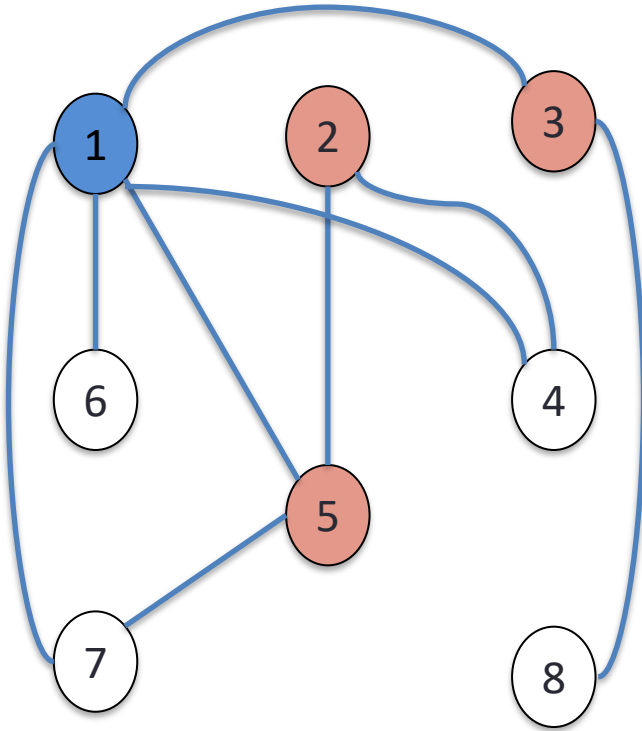
$$\neg x_j \equiv (1 - y_j)$$

Clauses

Inequalities

$$(x_1 \text{ OR } x_2 \text{ OR } \neg x_5) \rightarrow y_1 + y_2 + (1 - y_5) \geq 1$$

Vertex Cover Problem



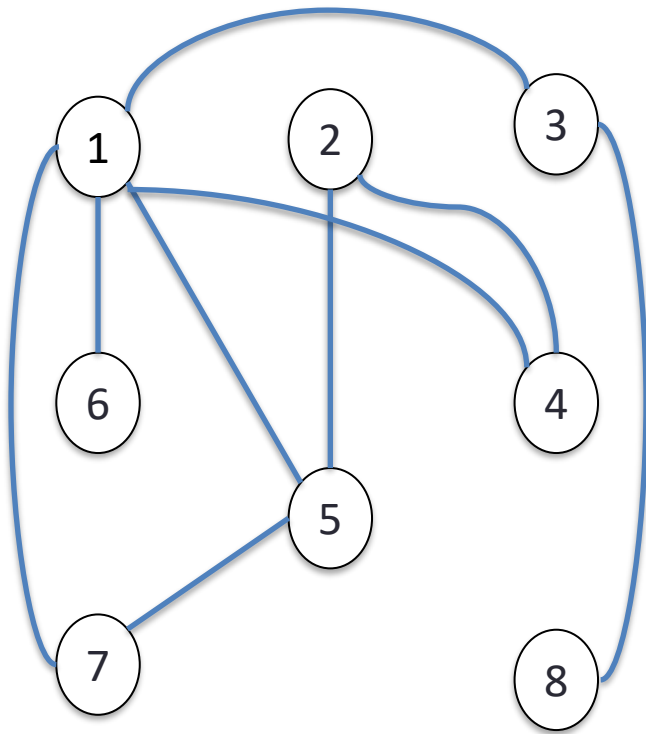
Choose smallest subset of vertices such that every edge must be “covered”

Eg, $\{1, 2, 3, 5\}$

or

$\{1, 2, 3, 7\}$

ILP for the vertex cover problem (Example)

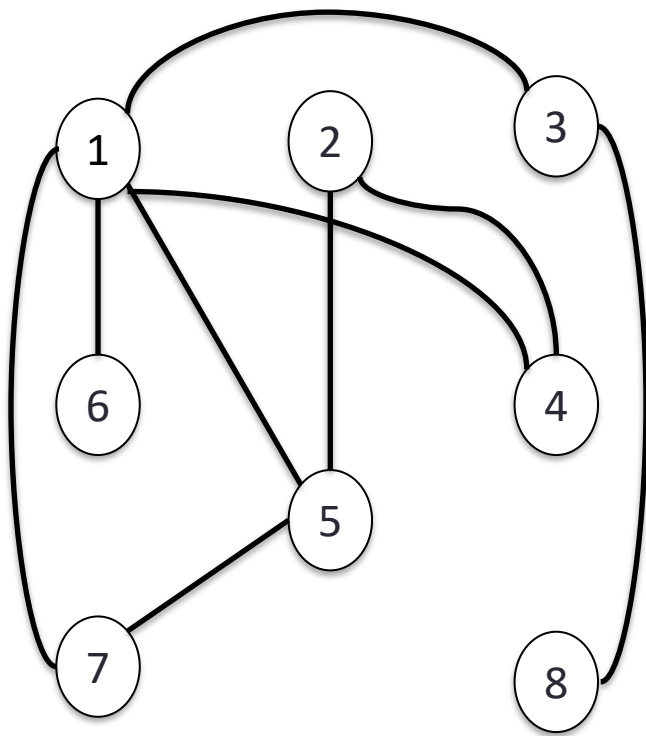


ILP decision variables

$$x_1, \dots, x_8$$

$$x_i = \begin{cases} 0 & \text{Vertex \# } i \text{ not chosen in subset} \\ 1 & \text{Vertex \# } i \text{ is chosen in subset} \end{cases}$$

ILP for the vertex cover problem (Example)



$$\begin{array}{llll} \min & x_1 + x_2 + \cdots + x_8 & & \\ \text{s.t.} & x_1 + x_7 \geq 1 & \leftarrow \text{Edge: (1,7)} & \\ & x_1 + x_6 \geq 1 & \leftarrow \text{Edge: (1,6)} & \\ & x_2 + x_4 \geq 1 & & \\ & \cdots & & \\ & x_i + x_j \geq 1 & \leftarrow (i,j) \in E & \\ & \cdots & & \\ & x_1 \leq 1 & & \\ & \vdots & & \\ & x_8 \leq 1 & & \\ & x_1, \dots, x_8 \geq 0 & & \\ & x_1, \dots, x_8 \in \mathbb{Z} & & \end{array}$$

Vertex Cover to ILP

- Vertices $\{1, \dots, n\}$
 - Decision variables:

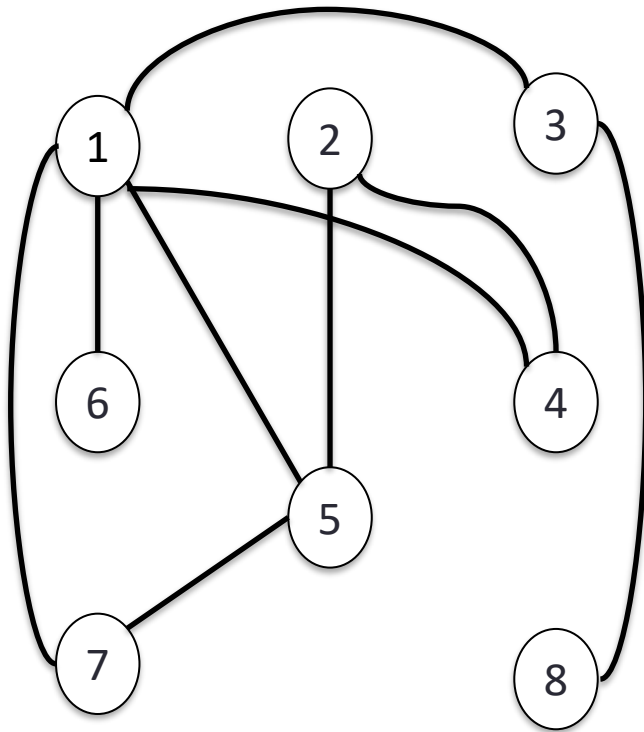
$$x_1, \dots, x_n$$

$$x_i \in \{0, 1\}$$

$$\begin{array}{ll} \min & \sum_{i=1}^n x_i \\ \text{s.t.} & 0 \leq x_i \leq 1 \quad \forall i \in V \\ & x_i + x_j \geq 1 \quad \forall (i, j) \in E \\ & x_i \in \mathbb{Z} \quad \forall i \in V \end{array}$$

LP relaxation of a vertex cover

- Problem: we may get fractional solution.



x_1	1
x_2	1
x_3	$\frac{3}{4}$
x_4	0
x_5	$\frac{5}{6}$
x_6	0
x_7	$\frac{1}{6}$
x_8	$\frac{1}{4}$

Objective value: 4

But solution meaningless for vertex cover.

Rounding Scheme

- Simple rounding scheme:

$$x_i^* \geq \frac{1}{2} \rightarrow x_i = 1$$

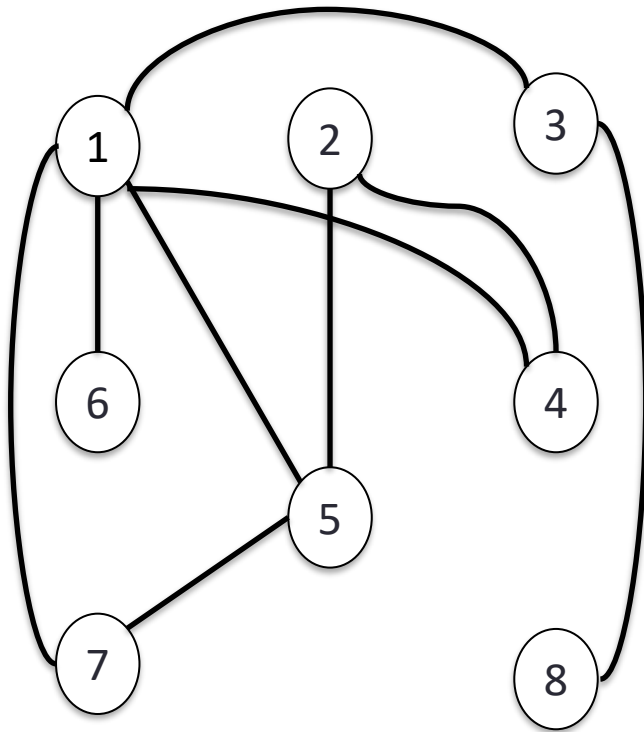
Real-Optimal Solution
is at least 0.5

Include vertex in
the cover.

$$x_i^* < \frac{1}{2} \rightarrow x_i = 0$$

LP relaxation of a vertex cover

- Problem: we may get fractional solution.



x_1	1
x_2	1
x_3	$\frac{3}{4}$
x_4	0
x_5	$\frac{5}{6}$
x_6	0
x_7	$\frac{1}{6}$
x_8	$\frac{1}{4}$



x_1	1
x_2	1
x_3	1
x_4	0
x_5	1
x_6	0
x_7	0
x_8	0

Rounding Scheme

Rounding scheme takes optimal fractional solution from LP relaxation and produces an integral solution.

$$\mathbf{x}^* \xrightarrow{\text{rounding}} \hat{\mathbf{x}}$$

1. Does rounding always produces a valid vertex cover?
2. How does the rounded solution compare to the opt. solution?

Rounding Scheme Produces a Cover

$$\mathbf{x}^* \xrightarrow{\text{rounding}} \hat{\mathbf{x}}$$

$$x_i^* + x_j^* \geq 1, \quad \text{for each } (i, j) \in E$$

$$\hat{x}_i = 1 \text{ or } \hat{x}_j = 1 \text{ for each } (i, j) \in E$$

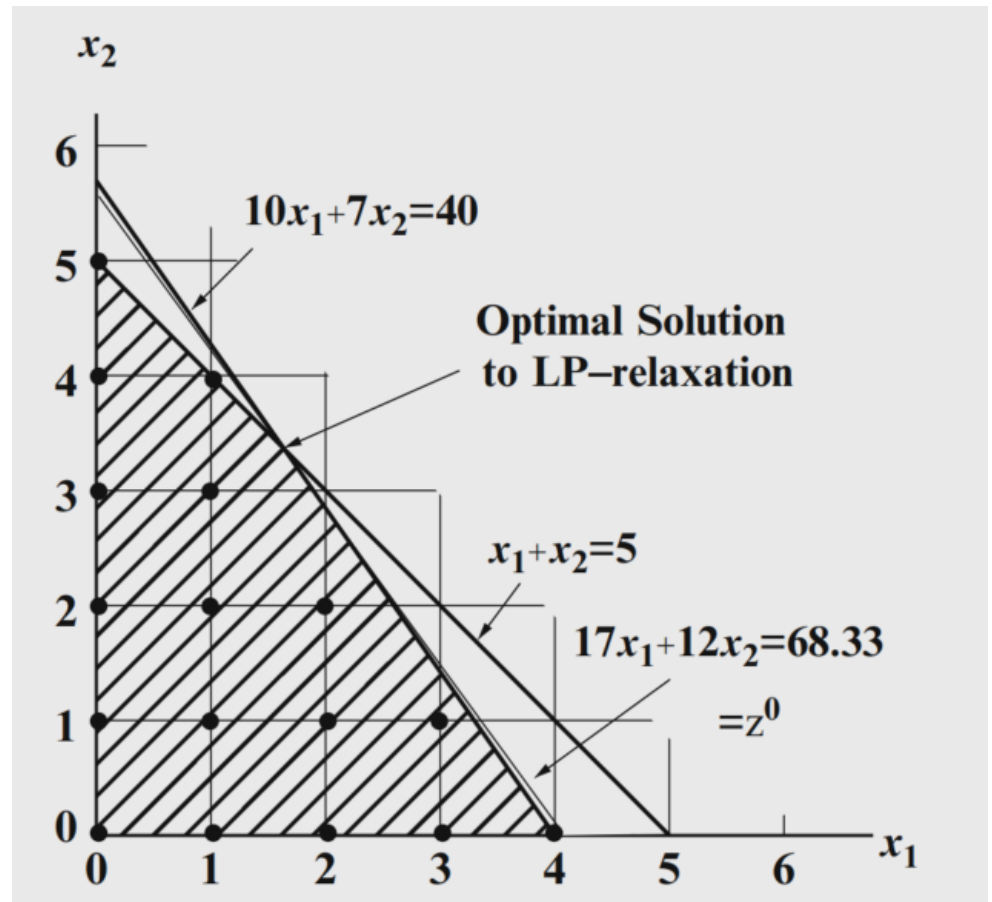
To Prove: The solution obtained after rounding covers every edge.

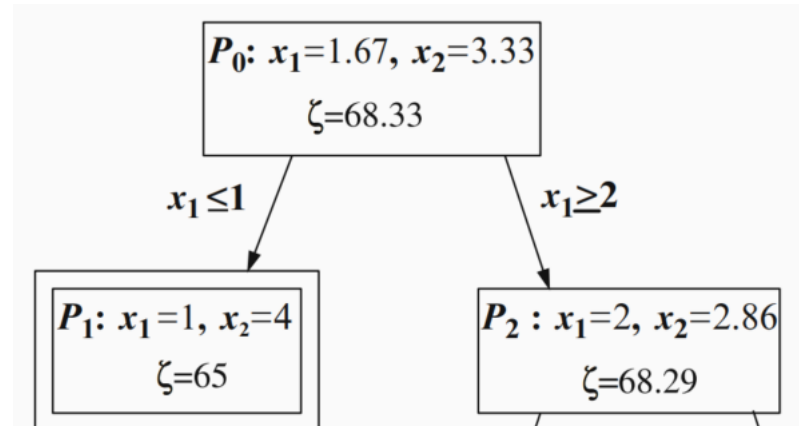
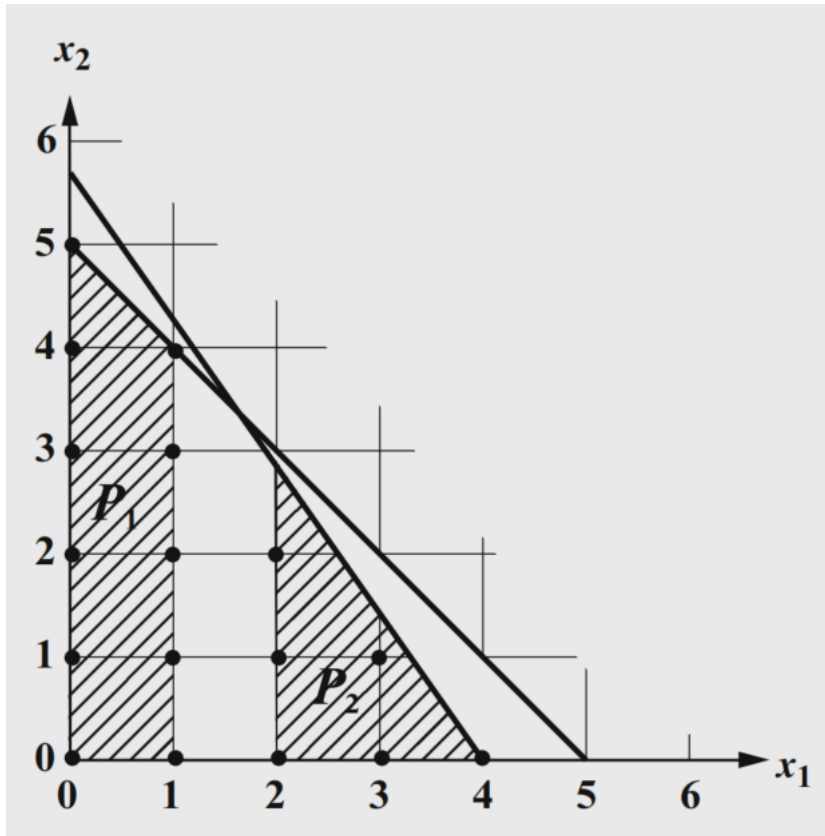
Approximation guarantee

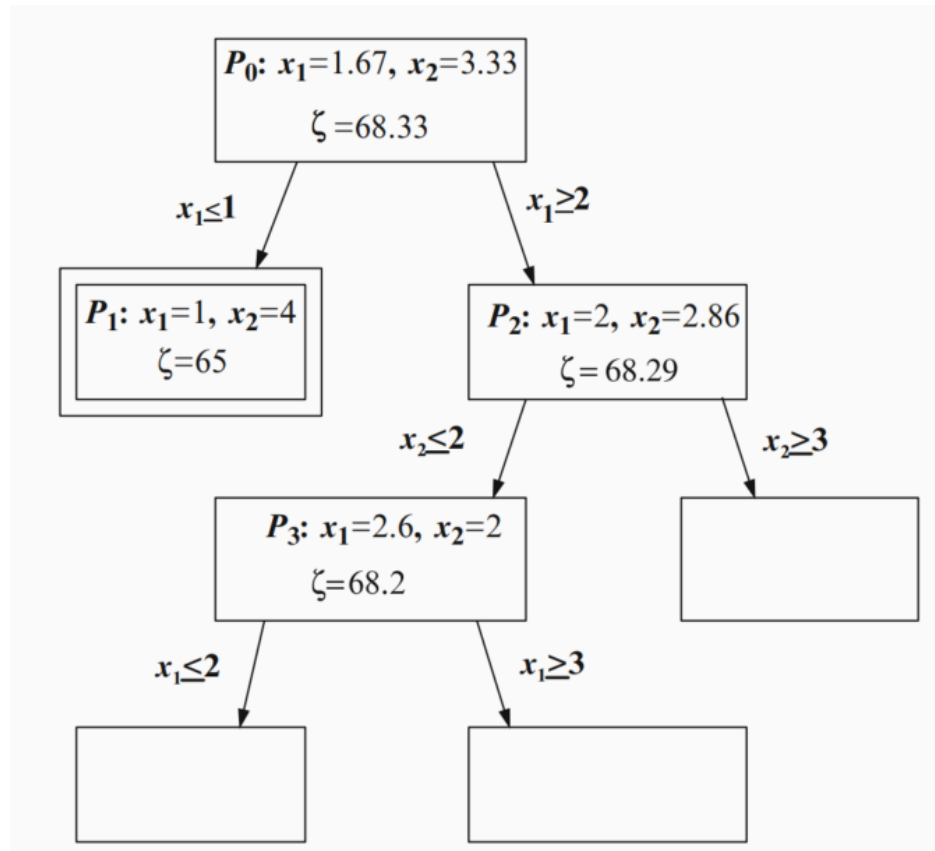
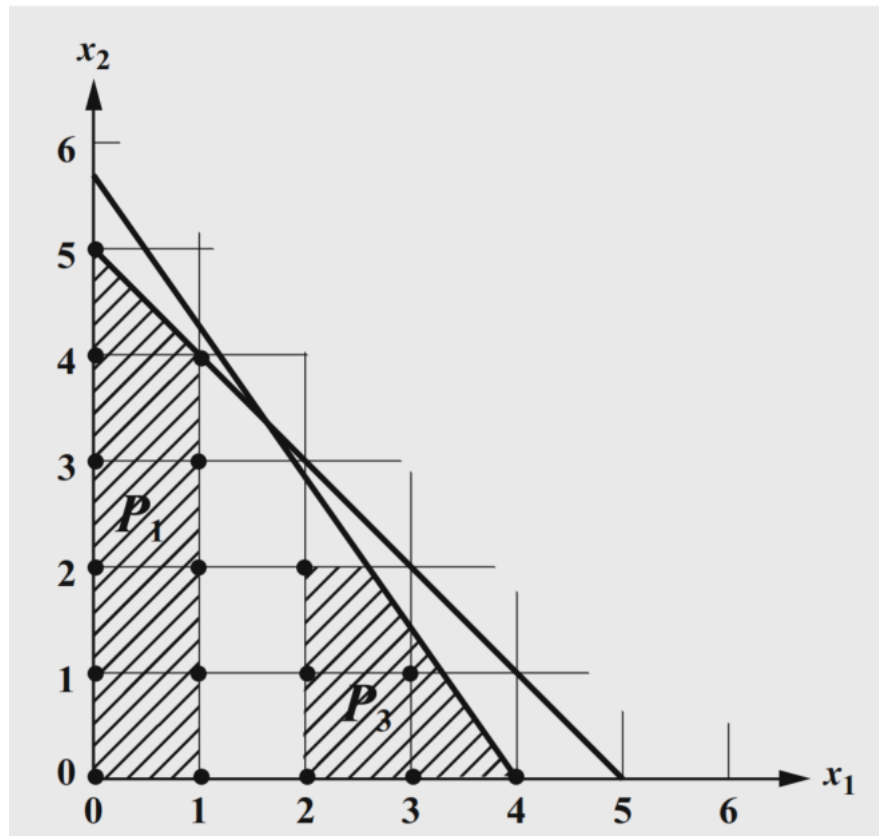
- **Theorem #1:** Rounding scheme yields a vertex cover.
- Cost of the solution obtained by rounding: C
- Optimal vertex cover cost: C^*
- **Theorem #2:** $C^* \leq C \leq 2 C^*$
- LP relaxation + rounding scheme:
 - 2-approximation for vertex cover!!

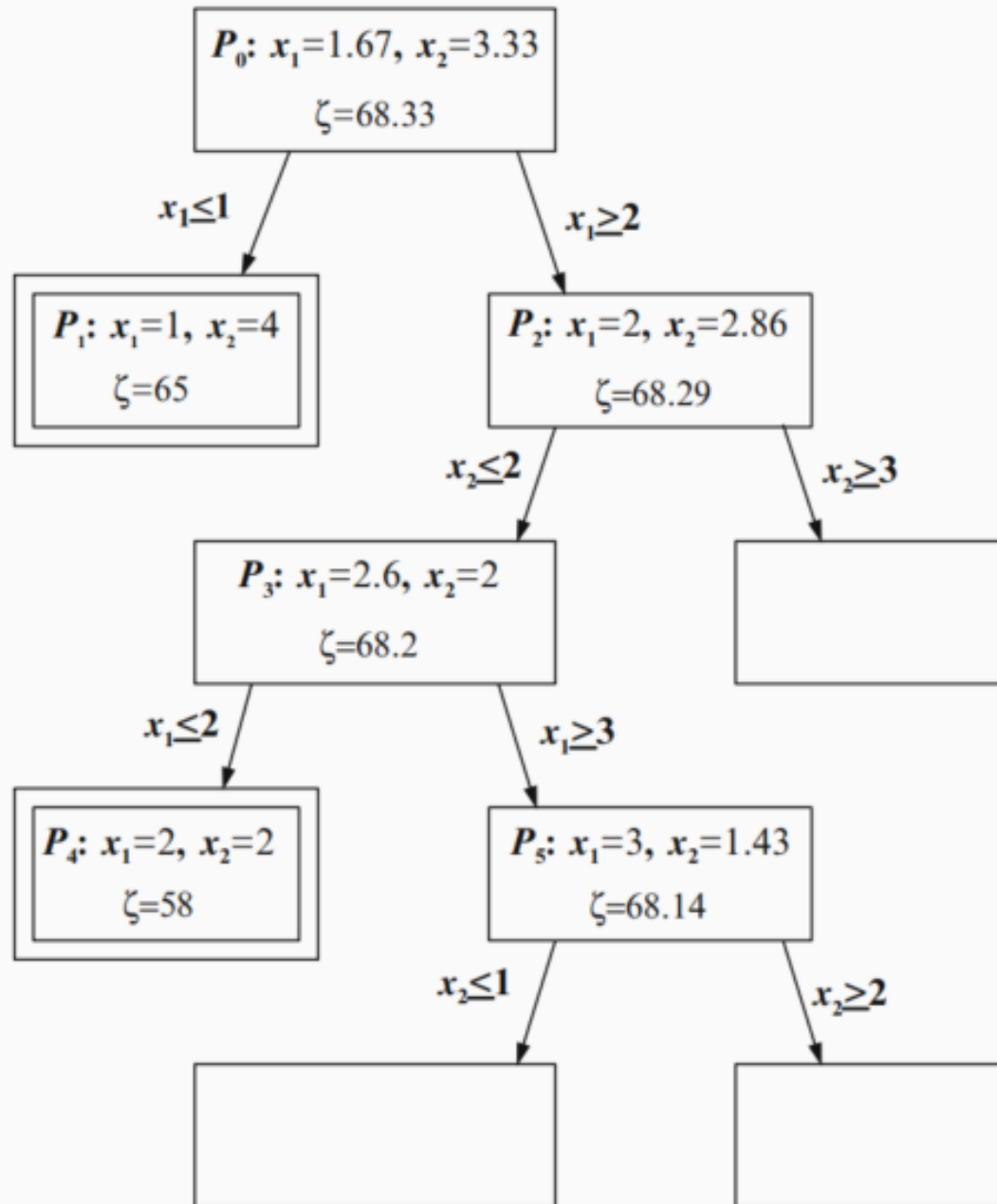
Branch-and-bound algorithm

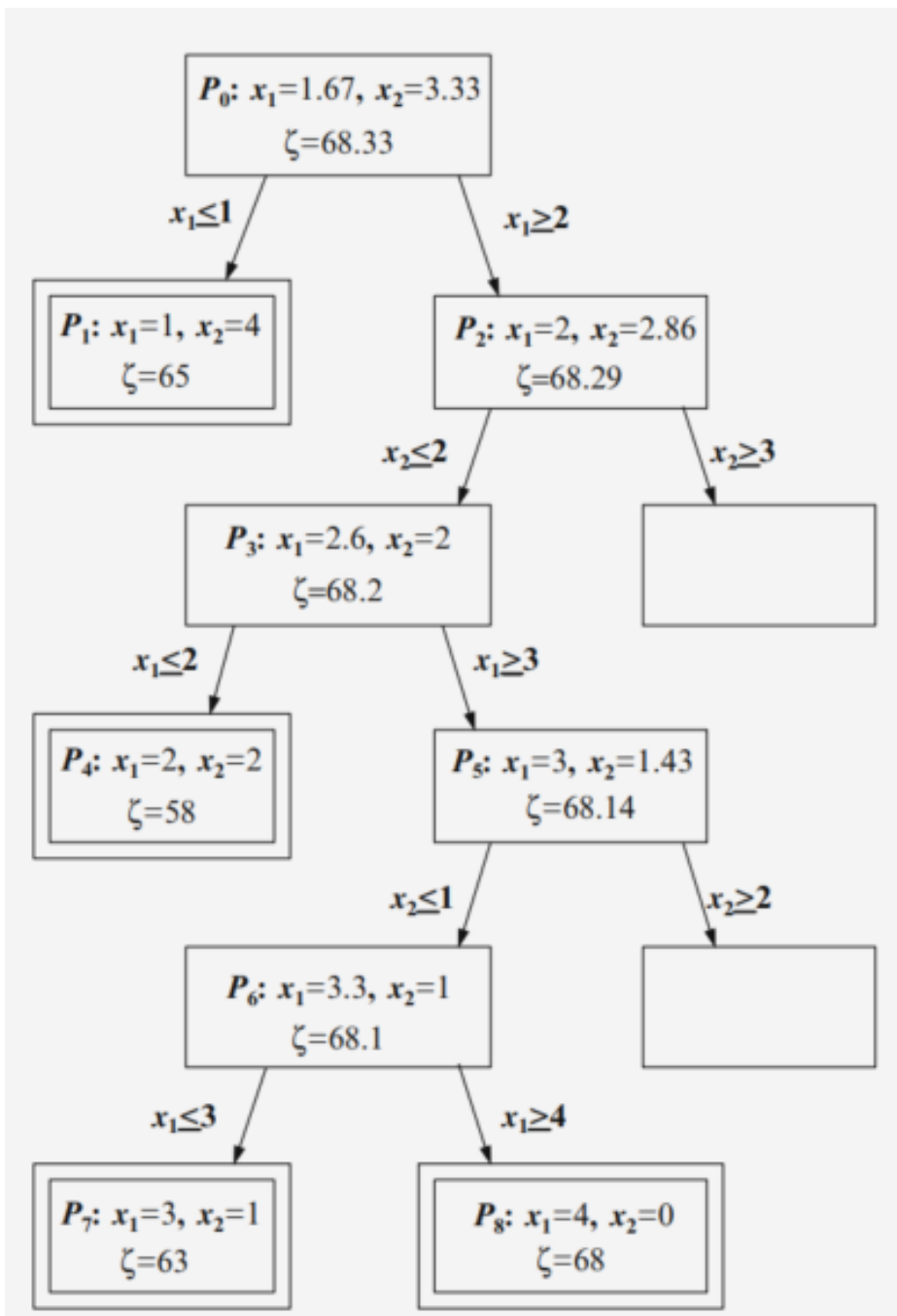
$$\begin{array}{llllll} \text{maximize} & 17x_1 & + & 12x_2 & & \\ \text{subject to} & 10x_1 & + & 7x_2 & \leq & 40 \\ & x_1 & + & x_2 & \leq & 5 \\ & & & x_1, x_2 & \geq & 0 \\ & & & x_1, x_2 & \text{integers.} & \end{array}$$











Summary

- Formulate a problem as an LP problem
- Applications of LP
- Duality of LP problem
- Sensitivity and parametric analysis
- Introduction to ILP

References

1. Robert J Vanderbei, Linear Programming: Foundations and Extensions
2. D. Bertsimas, Introduce to Linear Optimization

Q&A